Name: **Roll No:** **Batch:**

Pre-reading activity - Discuss the following questions in small groups or with a partner.

1. What are some of the most transformative technological innovations ever according to you?
2. Have all these innovations always been for the better?

Connecting technology and society: What students who want to build a career in technology that has a social impact should know by **Balaji Parthasarathy**

The Oxford dictionary defines technology as "the branch of knowledge dealing with Engineering or Applied Sciences" but it makes no reference to society. So, why bother building a career in technology that impacts society? For an answer, it is worth recalling late historian Marvin Kranzberg's laws of technology.

One states that "all history is relevant, but the history of technology is the most relevant". Another is "technology is a very human activity – and so is the history of technology." Between them, they make clear that human history can be understood by studying technological change, and technology is central to our activities as a human race. Thus, attempting to distinguish between technology and society is to seek a non-existent distinction.

If there is no distinction between technology and society i.e., if technology is not "outside" of society, how is the reference to impact meaningful? Here again, two other laws offer insights. One states that "technology is neither good nor bad; nor is it neutral." The other is that "although technology might be a prime element in many public issues, non-technical factors take precedence in technology-policy decisions." These suggest that even if technological discoveries and inventions are conceived by knowledge of Engineering or Applied Sciences, there are other influences on technological trajectories and outcomes. In other words, to build a socially impactful technological career, a sound grasp of Engineering and Applied Sciences is only the necessary condition. It is also important to understand three other aspects.

Who is impacted?

First, when seeking social impact, it is crucial to ask 'who in society?'. There are various social groups, marked by differences in affluence, age, gender, literacy, and physical disability, and their expectations of technology are hardly identical. Each one's clout to

influence technologies and their impact is also far from equal. For instance, firms with deep pockets, often determine what technologies to invest in and to adopt. Similarly, governments use regulation to determine the ease of access to technologies and their use. The growth of the Indian software industry is a case in point. Despite being the world's largest exporter of software services, little of the expertise was targeted at the less lucrative local markets. It was not until the government policies extended incentives that firms began to consider 'inclusive innovation' and technologies for the 'bottom of the pyramid'.

Second, a technology rarely is effective on its own; rather, its impact is contingent on complementary technologies and social conditions. Therefore, technologists seeking social impact must understand how the complements are faring. For instance, although the study of Artificial Intelligence (AI) originated in the 1950s, it has become the rage only in the past decade as increasingly powerful and affordable hardware has become available. Further, if AI now powers digital platforms such as Uber or Swiggy, it is only because smartphones have become ubiquitous among last-mile 'gig' workers who deliver the services. But combining both technologies to deliver low-cost services is possible, unfortunately, only because high levels of (un)der-employment, in a legal vacuum that affords no regulatory protection, forces workers to accept miserly wages.

Three, any presumption that the pursuit of new or improved technologies to address contemporary social challenges will not create unforeseen challenges is misplaced. Here, gaining a historical perspective is important. to appreciate how the past is littered with technological trajectories that have triggered new challenges, or have led to older challenges resurfacing. For instance, anxieties about cybercrime, privacy risks, and fake news, that accompany the communication revolution unleashed by the Internet are uncannily similar to those that accompanied the spread of the telegraph in the 19th century. Not surprisingly, the telegraph is dubbed the "Victorian Internet". Kranzberg could well have said: "the more technologies change, the more their impact remains the same."

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Answer the following questions by referring to the report.

A. Understanding the Report.

1. Do you think the use of technology is critical to our nature as human beings?

2. Why does the author stress the importance of understanding technology and society as co-constitutive rather than isolated entities?

3. Does the judicious use of technology largely define humans as social animals?

4. What are the three conditions that the author highlights to alert technological professionals regarding how technology and society both impact each other?

5. Do technological objects exist in a vacuum, or do they need to be part of a larger ecosystem to be effectively used?

B. Thinking Critically

1. Do you think the ease and convenience of delivery apps justify the problematic working conditions of delivery workers?

2. The report argues that technology does not only solve problems, but can also inadvertently create new problems. Name and describe one technological innovation that you feel has created this conundrum.

3. Do you think technology as a tool democratizes learning?